Differentials, Demystified

Why you *can* multiply by $\mathrm{d}x$ in a separable ODE, even though $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ is "not a fraction"

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The paradox, stated plainly. You have been told two things that seem to contradict each other:

(1) $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ is *not* a fraction. It is a single symbol for a limit, so you are not allowed to split it into a numerator $\mathrm{d}y$ and a denominator $\mathrm{d}x$.

(2) Yet to solve a separable equation you write $\mathrm{d}y = f(x)\,\mathrm{d}x$, integrate both sides, and it always gives the right answer.

Both statements are correct. The resolution is that step (2) is a **shorthand**: every "illegal" move is silently invoking a genuine theorem that happens to make the notation behave like a fraction. This document expands the shorthand back into the theorems it stands for.

1 First, why the warning is literally true

By definition,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x},$$

where $\Delta y = y(x + \Delta x) - y(x)$. The quotient $\Delta y/\Delta x$ *is* an honest fraction of two real numbers. But the derivative is the *limit* of that fraction, and a limit is a single number. There is no leftover "$\mathrm{d}y$" or "$\mathrm{d}x$" sitting around once the limit has been taken — they were consumed in the limiting process. So at the level of introductory calculus, $\mathrm{d}y/\mathrm{d}x$ is one indivisible symbol, and the warning is right.

The interesting question is therefore not "is it a fraction?" (it is not) but "*why does treating it like one keep working?*" The answer is a single theorem.

2 The one theorem behind all of it

Theorem 2.1 (Substitution rule, i.e. the chain rule run backwards)**.** *If $F' = f$ and u is a differentiable function of x, then*

$$\int f\big(u(x)\big)\, u'(x)\,\mathrm{d}x = F\big(u(x)\big) + C = \int f(u)\,\mathrm{d}u.$$

The famous mnemonic "$\mathrm{d}u = u'(x)\,\mathrm{d}x$" is nothing more than a *name* for this theorem. The $\mathrm{d}x$ does not cancel because of algebra; it cancels because the theorem says the two integrals are equal.

Everything that looks like fraction juggling — u-substitution, separable ODEs, even the chain rule written as $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\mathrm{d}y}{\mathrm{d}u}\dfrac{\mathrm{d}u}{\mathrm{d}x}$ — is this one fact wearing different clothes. Leibniz deliberately designed the symbols so that the theorem would *look* like cancellation. That is a feature, not an accident (more on this at the end).

A second, equivalent phrasing is worth keeping in your pocket, because it makes the ODE argument airtight:

Two functions of x with the same derivative differ only by a constant.

3 Separable ODEs: the shorthand, then the expansion

A separable equation has the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = g(x)\,h(y).$$

Shorthand (the way it is usually written). Treat $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ as a fraction and shuffle symbols:

$$\frac{\mathrm{d}y}{h(y)} = g(x)\,\mathrm{d}x \qquad \Longrightarrow \qquad \int \frac{\mathrm{d}y}{h(y)} = \int g(x)\,\mathrm{d}x.$$

Two genuinely forbidden moves were used: "multiplying by $\mathrm{d}x$" and "slapping $\int$ on a loose $\mathrm{d}y$."

Expanded (what the shorthand abbreviates). No symbol is ever split. Remember that y is an unknown *function* $y(x)$, and assume $h(y) \neq 0$ on the interval of interest.

Step 1. Divide both sides by $h(y(x))$ — legal, ordinary division of numbers:

$$\frac{1}{h(y(x))}\,\frac{\mathrm{d}y}{\mathrm{d}x} = g(x).$$

Step 2. Both sides are now ordinary functions of x. If they are equal, their antiderivatives in x are equal up to a constant:

$$\int \frac{1}{h(y(x))}\,\frac{\mathrm{d}y}{\mathrm{d}x}\,\mathrm{d}x = \int g(x)\,\mathrm{d}x.$$

Step 3. Apply the substitution rule to the left integral, with the inner function $u = y(x)$ and $f(y) = 1/h(y)$:

$$\int \underbrace{\frac{1}{h(y(x))}\,\frac{\mathrm{d}y}{\mathrm{d}x}\,\mathrm{d}x}_{\text{theorem: this equals } \int \frac{1}{h(y)}\,\mathrm{d}y} = \int \frac{1}{h(y)}\,\mathrm{d}y.$$

Combining the three steps gives *exactly* the shorthand's conclusion,

$$\int \frac{1}{h(y)}\,\mathrm{d}y = \int g(x)\,\mathrm{d}x,$$

but nothing was ever divided into "$\mathrm{d}y$ over $\mathrm{d}x$." The shorthand and the expansion produce identical work; the difference is only that the expansion names a theorem at each step instead of pretending the symbol is a fraction.

4 A worked example, side by side

Solve $\frac{\mathrm{d}y}{\mathrm{d}x} = x\,y$. Here $g(x) = x$ and $h(y) = y$.

Shorthand.

$$\begin{aligned}
\frac{\mathrm{d}y}{\mathrm{d}x} &= x\,y \\
\frac{1}{y}\,\mathrm{d}y &= x\,\mathrm{d}x \\
\int \frac{1}{y}\,\mathrm{d}y &= \int x\,\mathrm{d}x \\
\ln|y| &= \tfrac{x^2}{2} + C \\
y &= A\,e^{x^2/2}
\end{aligned}$$

In full detail. $y = y(x)$, and suppose $y \neq 0$:

$$\begin{aligned}
\frac{1}{y(x)}\,y'(x) &= x \\
\frac{\mathrm{d}}{\mathrm{d}x}\ln|y(x)| &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\tfrac{x^2}{2}\right) \\
\therefore\quad \ln|y(x)| &= \tfrac{x^2}{2} + C \\
y &= A\,e^{x^2/2}
\end{aligned}$$

In the rigorous column, the second line is the *whole trick*: by the chain rule $\dfrac{\mathrm{d}}{\mathrm{d}x}\ln|y(x)| = \dfrac{y'(x)}{y(x)}$, so the left side really is the derivative of $\ln|y|$. Once both sides are recognised as derivatives of known functions, "two functions with the same derivative differ by a constant" finishes the job. The shorthand's "$\int \frac{1}{y}\,\mathrm{d}y = \ln|y|$" was secretly this chain-rule recognition all along.

An honest footnote the shorthand hides: dividing by y in step one quietly assumed $y \neq 0$, so it *drops* the constant solution $y \equiv 0$. The expanded constant $A = \pm e^C$ ranges over nonzero values; allowing $A = 0$ afterwards restores the lost solution. The careful version makes such assumptions visible.

5 The same engine drives u-substitution

Evaluate $\displaystyle\int 2x\cos(x^2)\,\mathrm{d}x$. The shorthand sets $u = x^2$, writes "$\mathrm{d}u = 2x\,\mathrm{d}x$," and gets

$$\int \cos u\,\mathrm{d}u = \sin u + C = \sin(x^2) + C.$$

Why is "$\mathrm{d}u = 2x\,\mathrm{d}x$" allowed? Because it is the same substitution theorem, and you can verify the result with no differentials at all: by the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(x^2) = \cos(x^2)\cdot 2x,$$

which is exactly the integrand. The "$\mathrm{d}u$" bookkeeping is just a fast way to organise this chain-rule check.

6 Exact equations: the same trick in two variables

Consider a curve defined implicitly by $F(x, y) = C$, describable locally as $y = y(x)$. Differentiating both sides looks, on its face, like more forbidden fraction algebra:

$$F_x + F_y \frac{\mathrm{d}y}{\mathrm{d}x} = 0 \qquad \Longrightarrow \qquad F_x \,\mathrm{d}x + F_y \,\mathrm{d}y = 0,$$

where $F_x = \frac{\partial F}{\partial x}$ and $F_y = \frac{\partial F}{\partial y}$ are evaluated along the curve. The right-hand equation is exactly the form

$$M \,\mathrm{d}x + N \,\mathrm{d}y = 0, \qquad M = F_x,\ N = F_y,$$

that a differential-equations course calls an **exact equation**. The step from the implicit equation to this one again looks like cross-multiplying $\mathrm{d}x$ into the indivisible symbol $\dfrac{\mathrm{d}y}{\mathrm{d}x}$. What actually licenses it?

6.1 The honest derivation

Theorem 6.1 (Total derivative, i.e. the chain rule for two variables)**.** *Let $F(x, y)$ have continuous partial derivatives, and let $y = y(x)$ be differentiable. Then*

$$\frac{\mathrm{d}}{\mathrm{d}x} F\big(x, y(x)\big) \;=\; F_x\big(x, y(x)\big) + F_y\big(x, y(x)\big)\, y'(x).$$

This is proved the same way the ordinary chain rule is — by bounding the error in the linear approximation of F near a point and taking a limit — with no differential ever divided or multiplied on its own[1]. It is an honest statement about the single real-valued function $x \mapsto F(x, y(x))$.

Now suppose $F(x, y(x)) = C$ for all x in some interval — that is what "$F(x, y) = C$ defines $y(x)$" means. Differentiating the constant function gives 0, so by the theorem,

$$F_x + F_y\, y'(x) = 0$$

for every such x. This is Step 1 of the separable-ODE argument again, in the same disguise: one honest equation between functions of x, obtained with the chain rule and never by splitting $\mathrm{d}y/\mathrm{d}x$.

6.2 What "$M \,\mathrm{d}x + N \,\mathrm{d}y = 0$" really packages

Run the logic forward to see what the compact notation stands for. Define, for a function of two variables, its **total differential**

$$\mathrm{d}F \;:=\; F_x \,\mathrm{d}x + F_y \,\mathrm{d}y.$$

Just as $\mathrm{d}y = \frac{\mathrm{d}y}{\mathrm{d}x}\,\mathrm{d}x$ is a genuine identity between 1-forms (Section 8), $\mathrm{d}F$ is not built by dividing anything — it is a new object, a linear combination of the symbols $\mathrm{d}x$ and $\mathrm{d}y$, defined precisely so that its restriction to a curve $y = y(x)$ recovers the chain-rule expression above:

$$\mathrm{d}F\big|_{y=y(x)} \;=\; \big(F_x + F_y\, y'(x)\big)\,\mathrm{d}x.$$

So the sentence "$F_x \,\mathrm{d}x + F_y \,\mathrm{d}y = 0$" is shorthand for "the total derivative of F along the curve is zero," which by the theorem above and *equal derivatives differ by a constant* (applied to the identically-zero derivative) means $F(x, y(x))$ is constant — exactly the C we started with. Nothing was cross-multiplied; the notation was engineered so that packaging the chain rule as "$\mathrm{d}F = 0$" and unpacking it back to "$F = C$" are inverse operations.

[1] The full ε–δ argument is carried out in the companion note *Rigorous Derivation of the Total Derivative.*

6.3 Running it the other way: recovering F from M and N

The differential-equations use of this idea starts from the other end. One is handed

$$M(x, y)\,\mathrm{d}x + N(x, y)\,\mathrm{d}y = 0$$

with no F in sight, and asked to solve it. The equation deserves to be called *exact* precisely when some F exists with $F_x = M$ and $F_y = N$ — i.e., when $M\,\mathrm{d}x + N\,\mathrm{d}y$ really is $\mathrm{d}F$ for some function F, so the argument above applies and the solution curves are the level sets $F(x, y) = C$.

Proposition 6.2 (Exactness test). *If $M = F_x$ and $N = F_y$ for some twice continuously differentiable F, then*

$$\frac{\partial M}{\partial y} = F_{xy} = F_{yx} = \frac{\partial N}{\partial x},$$

by equality of mixed partial derivatives. Conversely, if $M_y = N_x$ throughout a simply connected region, such an F exists.

The forward direction is Clairaut's theorem in disguise — another "obvious" cancellation ($F_{xy} = F_{yx}$) that is really a nontrivial theorem about mixed partials, not a symbol-swap. The converse is the same existence fact behind Section 8's 1-forms: a 1-form with zero exterior derivative is locally exact (the Poincaré lemma). The condition $M_y - N_x = 0$ is precisely the statement that $\mathrm{d}(M\,\mathrm{d}x + N\,\mathrm{d}y) = 0$.

Worked example. Solve $(2xy + 3)\,\mathrm{d}x + (x^2 - 1)\,\mathrm{d}y = 0$. Here $M_y = 2x = N_x$, so the equation is exact. Integrate M in x, holding y fixed:

$$F(x, y) = \int (2xy + 3)\,\mathrm{d}x = x^2y + 3x + g(y).$$

Match F_y against N to pin down g:

$$F_y = x^2 + g'(y) \stackrel{!}{=} x^2 - 1 \quad\Longrightarrow\quad g'(y) = -1 \quad\Longrightarrow\quad g(y) = -y.$$

So $F(x, y) = x^2y + 3x - y$, and the implicit solution is

$$x^2y + 3x - y = C.$$

Differentiating this back with the ordinary and chain rules reproduces the original equation exactly — confirming that "cross-multiplying by $\mathrm{d}x$" and "taking d of both sides of $F = C$" are the same theorem viewed from opposite ends.

A caution. Do not confuse this additive total differential with the multiplicative cyclic relation of Section 7. $\mathrm{d}F = F_x\,\mathrm{d}x + F_y\,\mathrm{d}y$ is a *sum*, licensed by the chain rule for a function of two variables; $(\partial x/\partial y)(\partial y/\partial z)(\partial z/\partial x) = -1$ is a *product* of three different partials of three different two-variable functions implicitly linked by one relation. No amount of total-differential bookkeeping turns one into the other. Keeping straight *which* theorem is in play is, as always, the whole game.

7 Where the fraction picture genuinely fails

The cancellation only works for a *first* derivative of *one* variable. Push it further and it bites back:

Second derivatives don't cancel. The symbol $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ is not $(\mathrm{d}y)^2/(\mathrm{d}x)^2$; you cannot treat the "2"s as exponents and simplify. It is shorthand for $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\dfrac{\mathrm{d}y}{\mathrm{d}x}\right)$.

Partial derivatives don't cancel. For three quantities linked by one relation,

$$\left(\frac{\partial x}{\partial y}\right)\left(\frac{\partial y}{\partial z}\right)\left(\frac{\partial z}{\partial x}\right) = -1,$$

not $+1$. Naive "cancellation" gives the wrong sign — a famous trap in thermodynamics.

This is exactly why instructors warn you off the fraction picture: it is a reliable mnemonic in a narrow setting and a source of real errors outside it. Knowing *which theorem* you are using tells you precisely when the shortcut is safe.

8 The deeper resolution: differentials really are objects

There is a more advanced viewpoint in which $\mathrm{d}x$ and $\mathrm{d}y$ are not informal shorthand at all, but well-defined mathematical objects called **differential 1-forms**. In that framework, for a function y of a single variable x, the equation

$$\mathrm{d}y \;=\; \frac{\mathrm{d}y}{\mathrm{d}x}\,\mathrm{d}x$$

is a genuine, provable identity — a true statement about 1-forms, not a mnemonic. Integration $\int(\cdots)\,\mathrm{d}x$ becomes integration of forms, and the substitution rule becomes the natural "change of variables" for forms.

So Leibniz's notation was not merely lucky. It anticipated, by about two centuries, a rigorous theory in which the symbols mean something and the cancellation is a theorem. Introductory courses say "not a fraction" because that theory is out of reach at that stage — but the manipulations were never wrong, only unjustified *at that level.*

In one breath. $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ is a limit, not a fraction — so the warning is true. But every fraction-style manipulation in u-substitution and separable ODEs is a disguised application of the substitution rule (the chain rule backwards), together with "equal derivatives differ by a constant." Exact equations run the same trick with the multivariable chain rule: $F_x\,\mathrm{d}x + F_y\,\mathrm{d}y = 0$ is shorthand for $\mathrm{d}F = 0$, and $M_y = N_x$ is shorthand for "a potential function F exists," which is Clairaut's theorem and the Poincaré lemma in disguise. Leibniz's notation is engineered so those theorems *look* like cancellation, and at the level of differential 1-forms the cancellation becomes literally true. Trust the shortcut for first derivatives in one variable, and for total differentials built additively from partial derivatives; reach for the theorem the moment you hit higher derivatives or naively cancelled partials.